United States International University-Africa

COURSE CODE:	**APP 4080A**
COURSE TITLE:	**Collaborative (Team work in) Software Development**
SEMESTER:	**Fall 2024**
LECTURER:	**Prof. Frederick Ogore**
TASK	**Group 3 Assignment**
<u>Group Members</u> **NAME AND REG.NO.**	Yanet niguse 670530 Ndirangu Sandra 662866 Zake Nathan 665948 Gary Mbungo 663952

CHAPTER 1: INTRODUCTION

1.1 Background to the Study

SDG 2 seeks to end hunger, achieve food security, and promote sustainable agriculture through local, adaptive solutions. Despite advances in food production, many regions, especially in developing countries, continue to face high levels of malnutrition due to poverty, conflict, and limited resources. Children under five are particularly vulnerable. This project supports SDG 2 by using technology to empower caregivers with tools and guidance to improve child nutrition and reduce malnutrition in underserved communities.

1.2 Statement of the Problem

Undernutrition in children is a severe problem, particularly in Africa, where a large number of children have stunting, wasting, and underweight conditions. In addition to impairing their health, these disorders raise mortality rates and restrict their potential. Undernutrition in Africa is becoming an urgent issue as it continues to climb, even as other parts of the world are witnessing improvements. In order to better the health and future of these at-risk kids, this research attempts to investigate the causes and consequences of childhood undernutrition.

1.3 Study Objectives

Main Objective:

- To develop a comprehensive technological application that helps identify, monitor, and manage malnutrition in children under five in Kenya, following IEEE standards.

Specific Objectives:

- Collect and analyze child nutrition and health data from multiple sources.
- Provide real-time alerts and recommendations to caregivers and health workers.
- Create a platform for resource distribution and support for malnourished children. Evaluate the application's impact on reducing malnutrition rates.

Research Questions:

- What are the primary reasons behind malnutrition in Kenyan children under the age of five?
- In what ways may technology enhance the tracking and handling of undernourishment in children?
- What key features are necessary for an effective malnutrition-combatting application?
- How will this application impact malnutrition rates over time?

1.5 Significance

At 52.5% of child fatalities, undernutrition and related diseases are the cause, malnutrition is a key risk factor for morbidity and mortality in children. A vicious cycle of declining health is initiated when undernutrition lowers immune systems, making illnesses more severe. slowed growth, compromised cognitive development, and subpar performance in the classroom and at work are all consequences of inadequate nutrition during the crucial first 1,000 days of life. Millions of children globally suffer from wasting, obesity, and stunting, underscoring the urgency of taking quick action. To fight hunger, this research aims to address the underlying causes of undernutrition, make healthy food and essential health services accessible, and encourage societies that are healthier and more productive.

1.6 Limitations

Developing a mobile application to address hunger in remote areas presents several challenges. Access to technology can be limited in these locations, making it difficult to reach the intended users. Language and cultural barriers require the app to be culturally sensitive and available in local languages, which can be complex and resource-intensive. Limited healthcare infrastructure in remote areas may hinder the app's effectiveness without additional support for healthcare facilities and nutritional food access. Monitoring and evaluating the app's impact is challenging in rural settings. Ensuring sustainability requires ongoing funding, technical support, and updates. Additionally, user engagement may be difficult due to cultural or educational barriers, and managing sensitive health data requires robust privacy and security measures, which can be tough to implement in low-resource environments.

1.7 Assumptions

This project assumes several key factors will support its success. First, it expects that children, caregivers, and other participants will be available and willing to engage during the study and testing phases. It also presumes that all necessary equipment, such as mobile phones and computers, will be functional and accessible throughout the project. The project's scope and specifications are assumed to remain stable without significant changes. Stakeholders' needs, including those of caregivers, health professionals, and funders, will be adequately addressed. The project is expected to follow the proposed timeline, completing all phases on schedule. Additionally, the external political, economic, and social environment is assumed to remain stable and supportive. Finally, sufficient resources, including funding and personnel, are expected to be available to ensure the project's successful completion. These assumptions aim to facilitate the development of sustainable solutions to combat child undernutrition and improve public health outcomes.

CHAPTER 2: LITERATURE REVIEW

2.1 Existing Solutions

2.1.1 Nutrition Facts

This is a non-commercial information about nutrition and health, which was founded in 2011 by Dr. Michael Greger. NutritionFacts.org was developed for the sake of educating the public on subjects like disease prevention, weight management, and dietary advice, the website provides videos, articles, and blogs that are all backed by research (Greger, 2011). It's Goals and Principles is to advance evidence-based nutrition education and to support a plant-based diet as a means of preventing and treating chronic illnesses. Since it is a non-profit, information supplied is guaranteed to be objective and unaffected by business interests. This website prioritizes openness, accessibility, and scientific integrity because all of its content is available to the public for free.

2.1.1.1 Educational Value

NutritionFacts.org provides a rich resource for educational purposes. It is particularly strong in promoting plant-based diets for disease prevention, offering detailed explanations and references

to the latest peer-reviewed research. The platform excels in its ability to break down complex nutritional concepts into accessible formats, making it suitable for both the general public and professionals in the health sector.

2.1.1.2 Limitations

One limitation is its heavy focus on plant-based diets, which may not address all dietary approaches or cultural preferences. Furthermore, while the site offers extensive research references, the volume of information may be overwhelming for beginners or individuals without a strong scientific background.

2.1.2 MyFitnessPal

Overview of MyFitnessPal Launched in 2005 by Mike Lee, MyFitnessPal is one of the most widely used health-tracking apps in the world, offering tools to log food intake, exercise, and weight management. The platform has grown into a community of millions of users worldwide, providing both free and premium subscription options. The app's popularity stems from its large user base and integration with other fitness devices like Fitbit and Apple Health (MyFitnessPal, 2022). User Base and Reach MyFitnessPal boasts millions of users globally, making it a leading platform for tracking calories, exercise, and weight loss goals. Its widespread popularity is due in part to its social features, including community forums and support groups.

2.1.2.1 Availability and Accessibility Strengths:

MyFitnessPal is available on multiple platforms, including iOS, Android, and the web. It supports a broad range of devices and offers integration with fitness trackers, making it highly adaptable for users. Additionally, its user-friendly interface helps users easily track their food and exercise. Limitations: A key limitation is its dependence on user-generated data, which can lead to inaccuracies. Another issue is that its premium features are locked behind a paywall, limiting full functionality for low-income users. Furthermore, while the app is widely accessible, it requires internet connectivity for many of its core features, which may be limiting in rural or underserved regions.

2.1.2.2 Credibility and Trustworthiness Strengths:

MyFitnessPal's strength lies in its large database and ease of use. It integrates verified nutritional data and has partnerships with several health professionals. Limitations: The app is heavily reliant on user-generated data, which may lead to inaccuracies in nutrient tracking. This reliance on crowdsourced information can make it difficult for users to trust the data they input and receive.

2.1.2.3 Educational Value

The app offers users tools to monitor their daily caloric intake, macronutrient balance, and exercise, providing insights into how food choices impact overall health. However, it does not provide extensive educational resources or personalized dietary recommendations, limiting its use for those seeking in-depth nutrition advice.

2.1.3 Cronometer

An Overview of Cronometer Aaron Davidson developed Cronometer in 2005 as a personal tool to monitor his compliance with the CRON (Calorie Restriction with Optimal Nutrition) diet. It developed over time into a feature-rich nutrition tracking software that is utilized by more than 9 million individuals globally (Cronometer: Eat Smarter, Live Better, 2022). Dietitians, nutritionists, and health-conscious people all use Cronometer because of its reputation for accuracy and comprehensive nutritional data. With its headquarters located in Revelstoke, British Columbia, Canada, the platform has a global user base. Goals and Principles Cronometer seeks to offer precise and thorough nutrition information so that people can choose their diets with knowledge. It provides data that has been carefully selected from reliable sources, including the USDA, in order to promote openness, scientific correctness, and user empowerment.

2.1.3.1 Availability and Accessibility Strengths:

The app is available across multiple platforms (iOS, Android, web), with an intuitive interface and offline functionality that makes it accessible even in areas with poor internet connectivity. Limitations: While the app allows offline logging, features such as database updates require an internet connection. The reliance on digital devices also limits accessibility for users without smartphones or computers.

2.1.3.2 Credibility and Trustworthiness Strengths:

Cronometer's database is curated from verified sources, ensuring reliable nutritional information. The app is widely used by healthcare professionals, enhancing its credibility. Limitations: Despite rigorous curation, some entries are user-generated, which introduces the risk of inaccuracies. The presence of premium features may also affect user perception of its neutrality.

2.1.3.3 Educational Value

Cronometer provides detailed nutrient tracking, offering users insights into their intake of vitamins, minerals, and other micronutrients. However, the depth of information may be overwhelming for beginners. The app lacks personalized meal plans, which could otherwise improve user engagement.

2.1.3.4 Economic and Accessibility Considerations Strengths:

The app offers a free version, making it accessible to users who cannot afford premium tools. Its premium version is relatively affordable compared to similar apps. Limitations: The cost of premium features may deter low-income users from accessing advanced functionalities. The requirement of a smartphone or computer further limits accessibility, particularly in underserved areas.

2.1.3.5 Security and Privacy Strengths:

Cronometer offers robust data security measures to protect user privacy. Users retain control over their data and can decide what to share. Limitations: Like any digital platform, there is a risk of data breaches, which could compromise user information.

2.1.3.6 Impact on Addressing Nutritional Issues Strengths:

By offering detailed, accurate nutritional data, Cronometer enhances user awareness of dietary habits and fosters healthier choices. It integrates well with health professionals, making it a useful tool for personalized dietary monitoring.

Limitations: The app primarily caters to adults and lacks features tailored specifically to the nutritional needs of children, especially those in disadvantaged situations. Moreover, it may not always cater to diverse cultural dietary preferences. Comparative Analysis and Unique Value Proposition of the Proposed Solution While NutritionFacts.org, MyFitnessPal, and Cronometer

each contribute to the field of nutrition tracking and education, there are notable gaps that the proposed app aims to address.

Common Limitations Identified:

Internet Dependency: All platforms heavily rely on internet connectivity for features such as database updates, syncing, and premium features, which limits their usability in rural areas with limited internet access.

Cultural Relevance: The apps generally lack cultural adaptations for diverse populations, offering standardized dietary advice that may not reflect local food preferences or traditions. Personalized Health Advice: Although these platforms provide general dietary insights, they fall short in offering personalized meal plans or health advice tailored to individual needs.

Monitoring Long-Term Health Improvements: The platforms' capacity to measure users' long-term health improvements—particularly in children—is limited by their lack of tools for doing so. This makes it difficult to keep track of users' growth and general well-being over time. Particular Worth:

The Proposed Solution's Proposition The suggested app seeks to address these shortcomings by offering a more thorough, approachable, and culturally sensitive method of health management and nutritional education.

Important characteristics consist of:

Improved Accessibility: The software works offline, making it usable in places with spotty internet connectivity. Those who use this function in underserved and rural areas especially benefit.

Cultural Adaptation: It is relevant and useful for users from a variety of cultural backgrounds since it contains dietary recommendations and meal plans that are adapted to local food preferences and customs.

Personalized Meal Plans and Health Advice: The app offers individualized meal plans based on users' specific nutritional needs, preferences, and health goals, bridging the gap between general nutritional education and personalized health advice.

Tracking Long-term Health Improvements: It provides tools to monitor users' long-term health metrics, particularly for children, tracking their development over time. This feature ensures that users can observe and manage their progress toward healthier lifestyles.

Integration with Local Healthcare Services: By collaborating with local healthcare providers, the app ensures users receive personalized health support and advice, enhancing its real-world applicability.

Holistic Educational Approach: The app goes beyond nutrition, incorporating modules on healthy lifestyle choices, exercise, and mental well-being, promoting comprehensive health education.

CHAPTER 3: METHODOLOGY

3.1 Introduction:

This chapter details the methodology applied in this research aimed at combating malnutrition among children under five in Kenya, particularly focusing on underprivileged areas. The methodology encompasses the research design, targeted population, sample size, system development approach, systems analysis, design and development, testing and validation, result analysis, and ethical considerations.

3.2 Research Design:

The research design employs a mixed-methods approach, integrating both qualitative and quantitative techniques to thoroughly explore malnutrition among children under five in underprivileged communities.

Qualitative Research:

Qualitative research is pivotal in understanding social and cultural phenomena by capturing subjective experiences, beliefs, and practices. For this study, qualitative data is collected through: Interviews: Engaging caregivers and healthcare workers to gather in-depth insights on the challenges and perceptions surrounding child nutrition. Focus Groups: Facilitating group

discussions with community leaders and caregivers to capture communal perspectives on child malnutrition and potential solutions.

Quantitative Research:

Quantitative research provides statistical analysis, measuring the extent of malnutrition through: Surveys: Distributed to households to collect data on children's dietary intake, health status, and socio-economic conditions.

Data Analysis: Reviewing existing health records to measure the prevalence and impact of malnutrition, focusing on conditions like stunting and wasting.

Justification for Mixed-Methods Approach :

The mixed-methods approach is selected for its ability to provide a comprehensive understanding of malnutrition by blending qualitative insights with quantitative rigor. This approach ensures both depth and generalizability, guiding actionable interventions.

3.2.1 Targeted Population:

The study targets children under five years old, their caregivers, and healthcare workers in Kenya, especially in rural and underserved areas where malnutrition is most prevalent. The estimated affected population is around 2 million children, primarily suffering from stunting and wasting.

3.2.2 Sample Size:

A sample size of 50 participants will be selected through stratified random sampling, ensuring representation from various regions and demographics. This sample includes children, caregivers, and healthcare workers, providing a broad perspective on malnutrition.

3.3 Choice of Proposed System Development Methodology:

The **Agile software** development methodology will be employed in this project due to its flexibility, iterative approach, and emphasis on user feedback.

3.3.1 Justification for the Choice of Agile Methodology:

Flexibility and Adaptability: Agile accommodates adjustments based on evolving needs, ensuring the project remains relevant.

Continuous Quality Assurance: Agile's iterative testing identifies issues early, ensuring a high-quality end product.

Risk Reduction: By breaking the project into smaller tasks, Agile mitigates risks and allows easier adaptation to changes.

Prompt User Feedback: Agile integrates feedback loops, ensuring the solution meets user needs.
Results-Oriented: Agile focuses on delivering functional outcomes rather than rigidly following processes, driving efficient project execution.

Solution: The NourishKids Platform

NourishKids is a collaborative project developed by a skilled team committed to addressing malnutrition. The solution starts with a web application, followed by a mobile app with offline capabilities to reach caregivers in rural areas. The platform offers a range of features designed to support and simplify child nutrition management:

Key Features:

1. Personalized Meal Plans:
   Users can generate tailored meal plans based on their child's specific nutritional requirements and the local food available. These plans can be accessed offline.

2. Nutrient Calculator:
   Caregivers can calculate their child's daily nutrient needs and ensure they receive balanced meals.

3. Predictive Nutritional Analysis:
   Based on growth patterns and dietary data, the platform forecasts future nutritional requirements, allowing for proactive planning.

4. Behavior Analysis:
   Tracking eating habits enables the platform to offer personalized suggestions for

overcoming challenges like picky eating. Caregivers also receive actionable tips on improving their child's diet.

5. Offline Access:
   The platform's resources, including meal plans and nutritional guides, will be stored locally on devices for offline access in areas with limited internet.

6. Dual-Platform Accessibility:
   NourishKids will be available on both web and mobile platforms, making it accessible to caregivers regardless of their device or internet access.

Roles and Responsibilities of the Team

This project is a collaborative effort, with each team member playing a crucial role in ensuring its success:

- Project Manager: Ndirangu Sandra
  Responsible for overseeing project operations, team coordination, and ensuring milestones are met on time.

- Developer: Yanet Niguse
  Leads the technical development of the *NourishKids* platform using the React framework for frontend development, and ensures that the web and mobile apps function smoothly.

- UI/UX Designer: Gary Mbungo
  Designs the user interface and experience, focusing on creating an intuitive and accessible platform for caregivers with varying degrees of technological literacy.

Note: The team is flexible and will collaborate. So the roles of the individual is not restricted

Technologies to be used:

- React for the web app
- Flutter for mobile app
- Github for collaborative work
- Figma for User Interface design
- Draw.io for designing and Modeling the project

- Firebase for storage

NourishKids Web App Development Timeline (Sept 19 - Dec 10, 2024)

- **Sept 19 - Sept 25**:
    - Project kickoff: finalize scope, assign roles, and set up tech stack. Begin wireframing and initial design ideas.
- **Sept 26 - Oct 2**:
    - Complete UI/UX design. Prepare the React development environment and backend architecture.
- **Oct 3 - Oct 9**:
    - Start development: build basic frontend structure (homepage, login, and dashboard).
- **Oct 10 - Oct 16**:
    - Develop core features like nutrient tracking, child profiles, and backend database setup.
- **Oct 17 - Oct 23**:
    - Implement additional features (nutrient recommendations, progress tracking). Integrate frontend with backend.
- **Oct 24 - Oct 30**:
    - Conduct internal testing, fix bugs, and refine features. Improve UI/UX based on testing feedback.
- **Oct 31 - Nov 6**:
    - Continue testing, bug fixing, and optimizing app performance. Ensure mobile and desktop responsiveness.
- **Nov 7 - Nov 13**:
    - Polish remaining features and fine-tune the design. Prepare for deployment.

- **Nov 14 - Nov 20**:
    - Final testing phase: fix final bugs and ensure a smooth user experience.
- **Nov 21 - Nov 27**:
    - Launch the web app, deploy it on hosting platforms, and initiate promotional activities.
- **Nov 28 - Dec 4**:
    - Post-launch monitoring: gather user feedback and fix any issues that arise.
- **Dec 5 - Dec 10**:
    - Final optimizations, performance improvements, and stakeholder review.

3.4 System Analysis and Design

3.4.1 Requirements Gathering and Analysis:

To effectively combat child undernutrition, the system's functional and non-functional requirements must be thoroughly understood.

3.4.1.1 Functional Requirements

Data Collection: The system must allow for the input of children's nutritional data by caregivers and health workers.

Real-Time Alerts: The app should provide alerts and recommendations based on the nutritional data to help caregivers act quickly.

Resource Allocation: Facilitate the distribution of resources like nutritional supplements for malnourished children.

Impact Evaluation: Monitor and evaluate the long-term impact of interventions on malnutrition rates.

3.4.1.2 Non-Functional Requirements

Performance: The app must respond quickly to ensure real-time monitoring.

Security: Protect sensitive health data, adhering to data protection laws. Usability: Ensure ease of use, even for users with limited technical skills.

Scalability: Allow the system to expand as more users and regions are added. Reliability: The system must be stable, with minimal downtime.

Use Case: Register Profile

Use Case ID	UC1
Use Case Name	Register Profile
Actor	Caregiver
Description	Allows a caregiver to register a new profile for a child.
Preconditions	The caregiver must be authenticated in the system.
Preconditions	A new profile for a child is created and stored in the system.
Main Flow	1. The caregiver selects the "Register Profile" option. 2. The system prompts for child details. 3. The caregiver enters the required details. 4. The system validates and saves the profile information. 5. A confirmation message is displayed to the caregiver.
Alternate Flows	4a. If validation fails, the system displays an error message and prompts for correct information.

Use Case: Manage Recommendation

Use Case ID	UC2
Use Case Name	Manage Profile
Actor	Caregiver
Description	Allows a caregiver to update or delete an existing profile.
Preconditions	The caregiver must be authenticated in the system and have existing profiles.
Preconditions	The child's profile is updated or deleted as requested by the caregiver.
Main Flow	1. The caregiver selects the "Manage Profile" option. 2. The system displays a list of profiles. 3. The caregiver selects a profile to update or delete. 4. The caregiver makes the desired changes. 5. The system validates and updates/deletes the profile. 6. A confirmation message is displayed to the caregiver.
Alternate Flows	5a. If validation fails, the system displays an error message and prompts for correct information.

Use Case: Provide Recommendation

Use Case ID	UC3
Use Case Name	Provide Recommendation
Actor	System
Description	The system provides meal and nutritional recommendations based on the child's profile and symptoms.
Preconditions	The child’s profile and health data are up-to-date.
Preconditions	Relevant recommendations are displayed to the caregiver
Main Flow	1. The system analyzes the child’s profile and health data. 2. The system generates appropriate recommendations. 3. The recommendations are displayed to the caregiver.
Alternate Flows	None.

Use Case: Nutrient Information

Use Case ID	UC4
Use Case Name	Nutrient Information
Actor	System

Description	The system retrieves and displays nutritional information for various food items
Preconditions	The system has access to the nutrition database.
Preconditions	Nutritional information is displayed to the caregiver.
Main Flow	1. The caregiver requests nutritional information. 2. The system retrieves data from the nutrition database. 3. The nutritional information is displayed to the caregiver.
Alternate Flows	None.

Use Case: Progress Tracking

Use Case ID	UC5
Use Case Name	Progress Tracking
Actor	Caregiver
Description	Allows a caregiver to track and view a child's progress over time.
Preconditions	The caregiver must be authenticated in the system and have existing profiles.
Preconditions	The child's progress data is displayed to the caregiver.

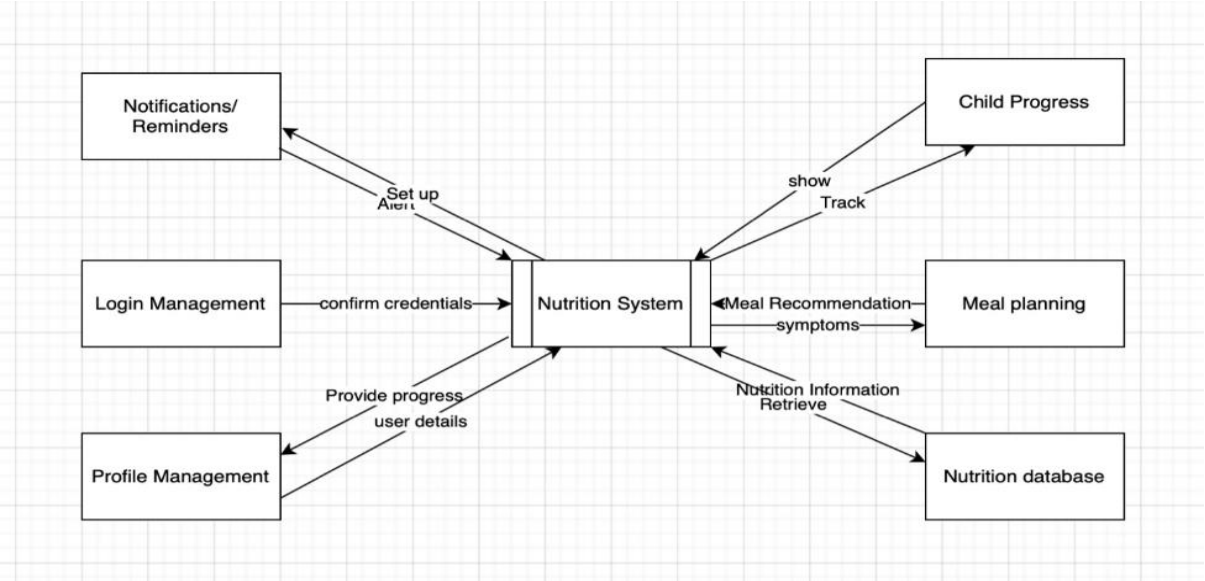
Main Flow	1. The caregiver selects the "Progress Tracking" option. 2. The system displays progress data for the selected child. 3. The caregiver views the progress data.
Alternate Flows	None.

Use Case: Send Alert

Use Case ID	UC6
Use Case Name	Send Alert
Actor	System
Description	The system sends alerts or reminders to the caregiver based on certain conditions.
Preconditions	The system has the caregiver's contact information and the conditions for alerts are met.
Preconditions	The caregiver receives the alert or reminder.

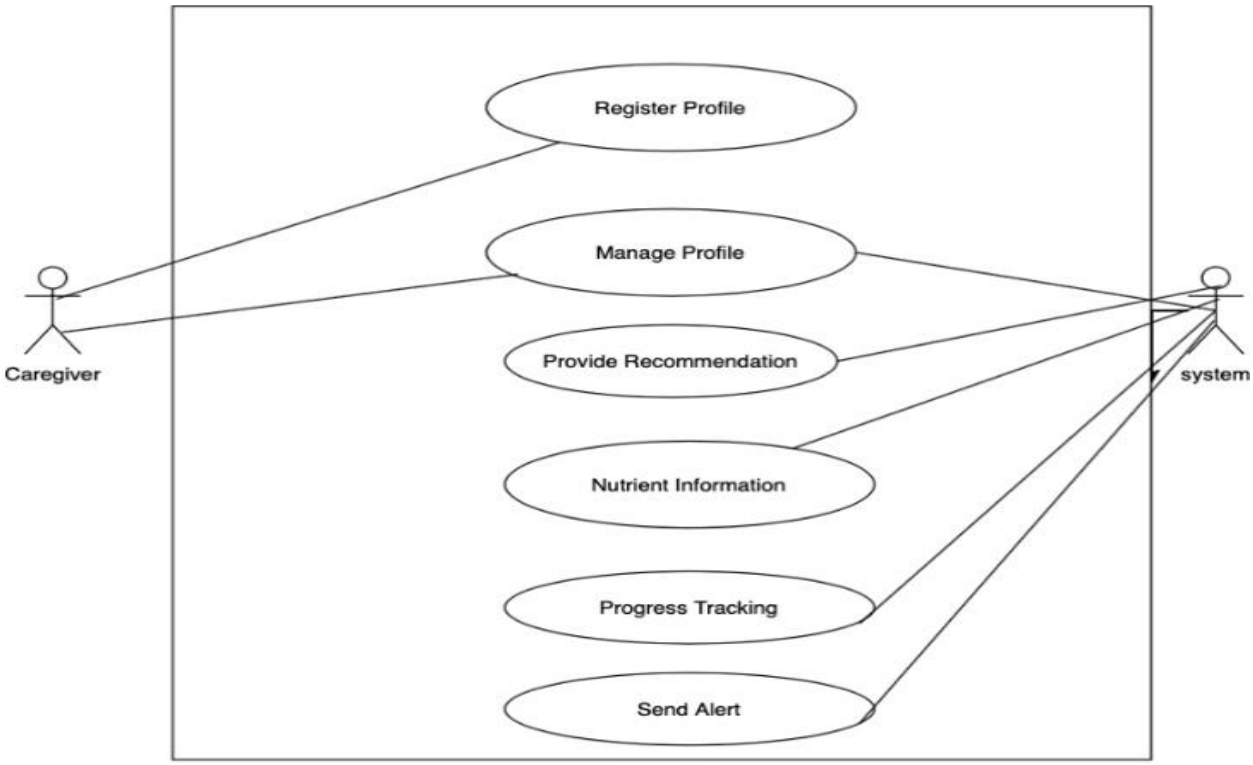
Main Flow	1. The system monitors the conditions for alerts. 2. When conditions are met, the system generates an alert. 3. The alert is sent to the caregiver. 4. The caregiver receives the alert.
Alternate Flows	None.

Dataflow Diagram

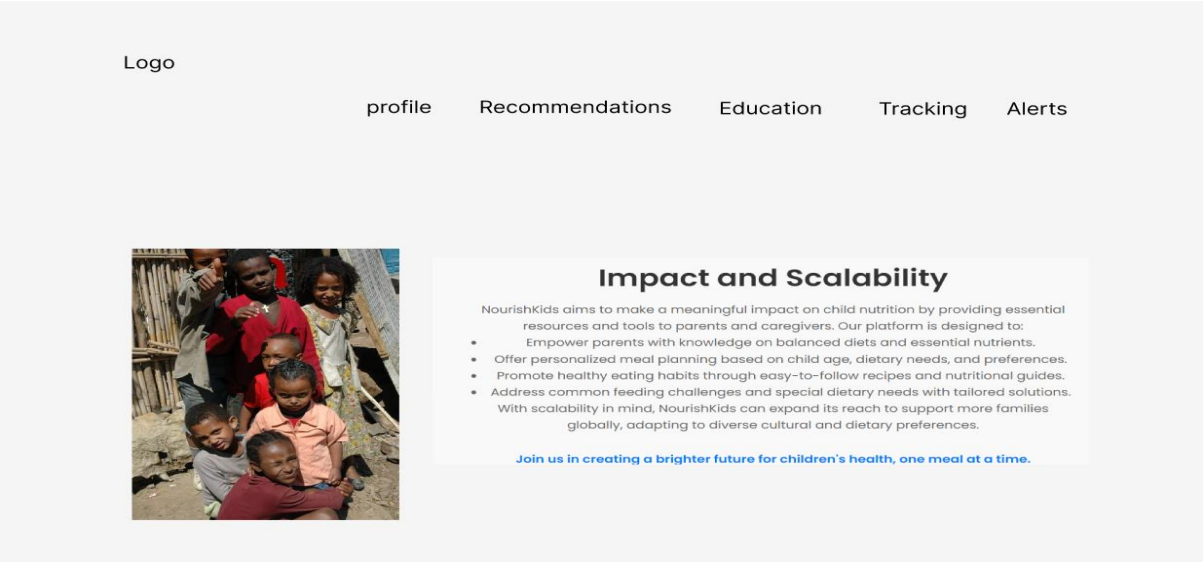


Use case diagram

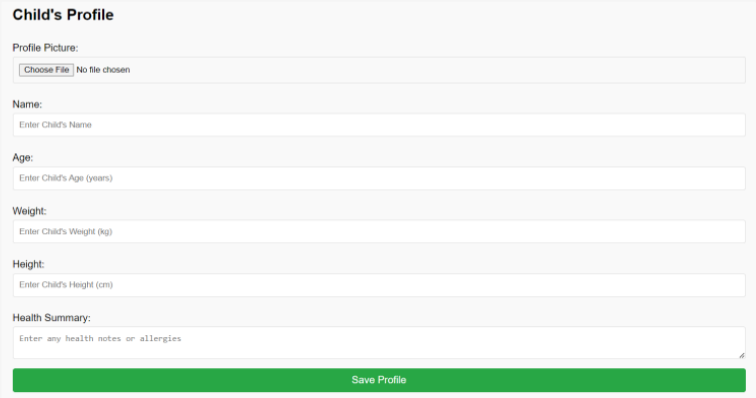
A use-case model describes a system's functional requirements in terms of use cases. It is a model of the system's intended functionality (use cases) and its environment (actors). Use cases 19 enable you to relate what you need from a system to how the system delivers on those needs (Gemino & Parker, 2009). The use case diagram for this proposed system is as follows: 3.4.4.2 Use case diagram A use-case model describes a system's functional requirements in terms of use cases. It is a model of the system's intended functionality (use cases) and its environment (actors). Use cases 19 enable you to relate what you need from a system to how the system delivers on those needs (Gemino & Parker, 2009). The use case diagram for this proposed system is as follows:



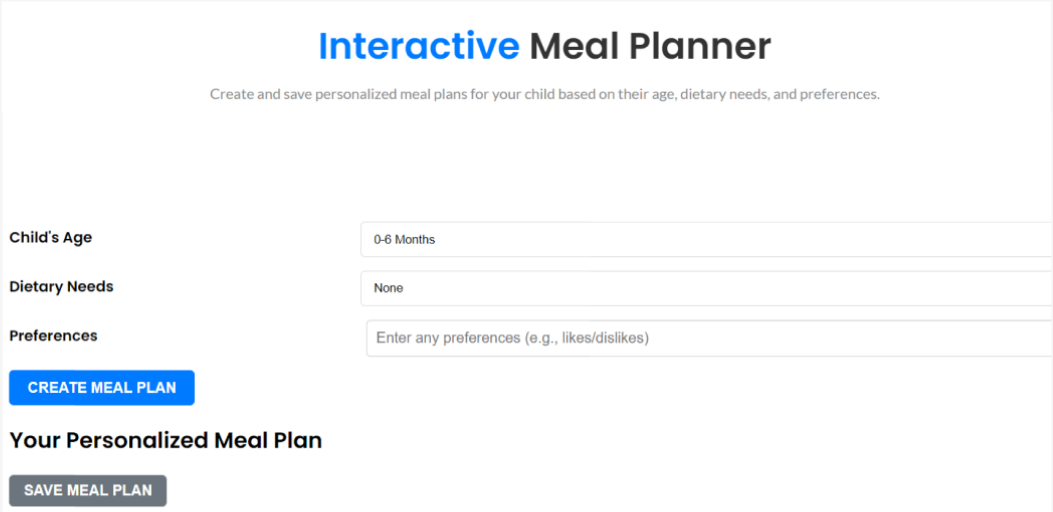
User Interface Mockups



Child's Profile Section



Recommended Meal Pan



Education

Nutritional Guides

Explore detailed guides on balanced diets, essential nutrients, and best feeding practices for young children.

Balanced Diets

A balanced diet is crucial for the growth and development of young children. Ensure a mix of proteins, carbohydrates, fats, vitamins, and minerals in their daily meals.

Essential Nutrients

Learn about the essential nutrients such as calcium, iron, and vitamins that are vital for your child's development.

Feeding Practices

Adopt best practices for feeding your child, including portion control, meal timing, and introducing new foods.

Meal Plans

Find sample meal plans tailored to different age groups and dietary needs.

0-6 Months

Exclusive breastfeeding or formula feeding. Gradual introduction of solid foods after 6 months.

6-12 Months

Introduce pureed fruits and vegetables, cereals, and lean proteins. Continue breastfeeding or formula feeding.

1-2 Years

Transition to more solid foods, including small pieces of fruits, vegetables, grains, and proteins. Ensure adequate hydration.

2-5 Years

Provide a variety of foods from all food groups. Focus on whole grains, lean proteins, fruits, and vegetables.

Tracking

Progress Summary

Growth Charts

Milestones Achieved

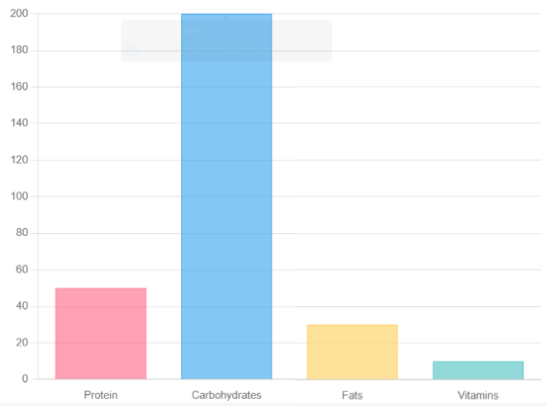
Milestones: Enter Milestones

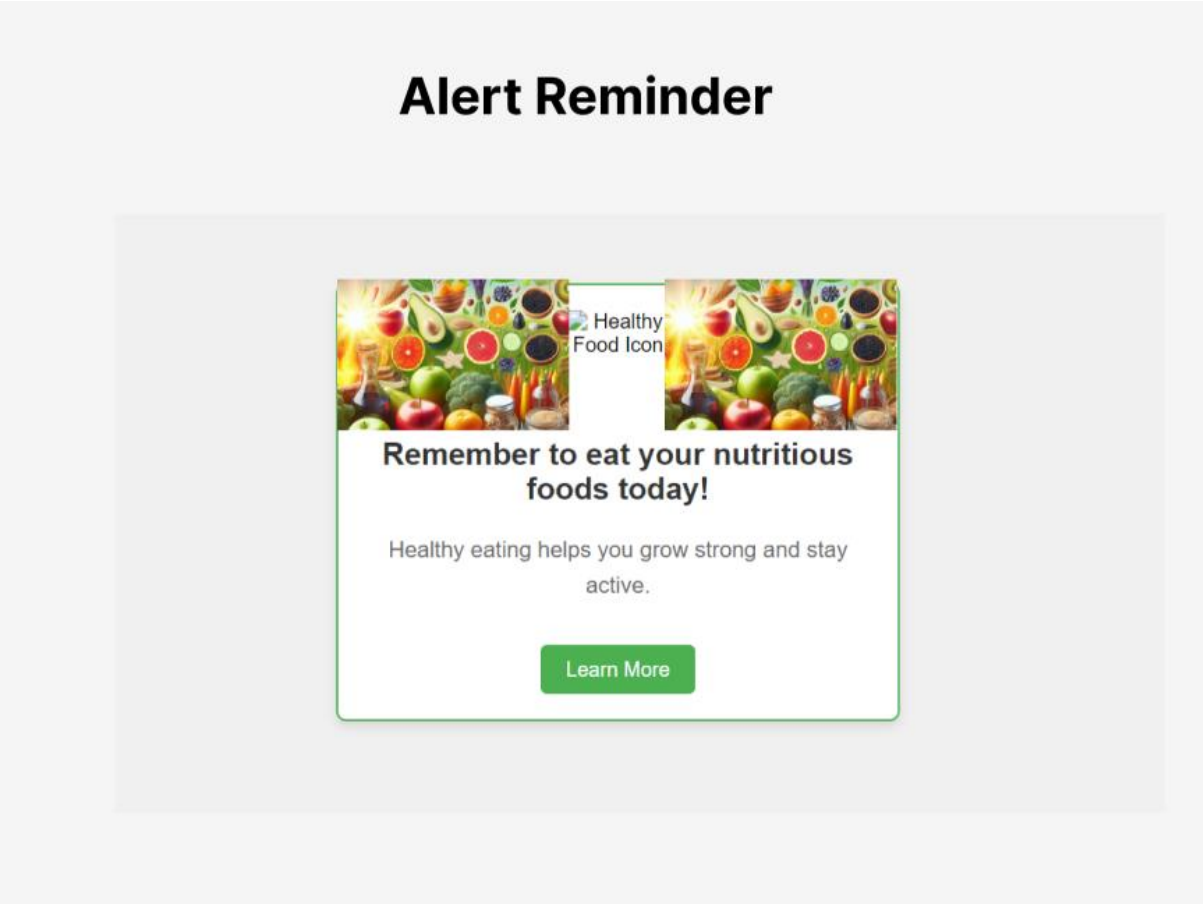
Weekly Progress

Height: Enter Child's Height (cm) Weight: Enter Child's Weight (kg)

Save Progress

Nutritional Data Visualization





CHAPTER 4: CONCLUSION AND RECOMMENDATION

The Nutrition System initiative, which attempts to address childhood malnutrition, has drawn attention to the innovative role that technology may play in managing health, particularly in helping caregivers improve their charges' nutritional condition. By including features like personalized meal planning, data management, and real-time alerts, the system offers a comprehensive strategy to address undernutrition in kids. After addressing issues with data processing, user interface design, and accuracy, a strong application was produced. The system should prioritize data protection, advanced analytics, scalability, user interface advancements, and interaction with healthcare providers in the future. The system can help fight malnutrition and accomplish Sustainable Development Goal 2 (Zero Hunger) by implementing these suggestions.

Reference

(n.d.). Calorie Tracker & BMR Calculator to Reach Your Goals | MyFitnessPal. https://www.myfitnesspal.com/

(2022, 21). Cronometer: Eat smarter. Live better. https://cronometer.com/

(n.d.). Economist Impact. https://impact.economist.com/perspectives/sites/default/files/economist_impact_breaking_the_cycle_of_chronic_malnutrition_in_ssa_feb_23.pdf

Impact of malnutrition on education inequality in Africa: A partial wavelet coherence analysis | Raimi | South African Journal of Economic and Management Sciences. (n.d.). South African Journal of Economic and Management Sciences. https://sajems.org/index.php/sajems/article/view/5268/2980

Malnutrition in children. (2023, June 27). UNICEF DATA. https://data.unicef.org/topic/nutrition/malnutrition/

(2012, November 8). NutritionFacts.org. https://nutritionfacts.org

The cost of malnutrition. (2021, August 25). Nation. https://nation.africa/kenya/healthy-nation/the-cost-of-malnutrition-3478484